

Importance-sampling Cross-validators Z-residuals for Diagnosing Bayesian Spatial Models

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Section 1

Motivation with A Bayesian Spatial Model for Disease Mapping

Scottish Lip Cancer Data

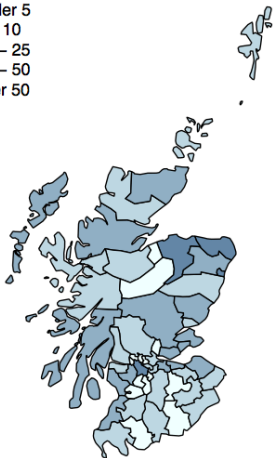
The data represents male lip cancer counts (over the period 1975 - 1980) in the $n = 56$ districts of Scotland. The data includes these columns:

- the number of observed cases of lip cancer, y_i ;
- the number of expected cases, E_i , which are based on age effects, and are proportional to a “population at risk” after such effects have been taken into account;
- the percent of the population employed in agriculture, fishing and forestry, X_i , used as a covariate; and
- a list of the neighbouring regions.

Scottish Lip Cancer Data (continued)

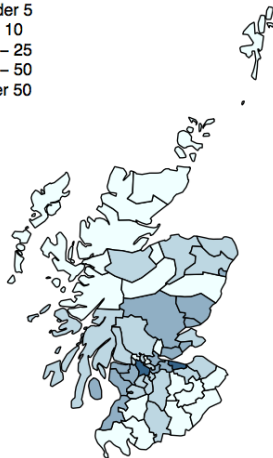
Observed

- under 5
- 5 – 10
- 10 – 25
- 25 – 50
- over 50



Expected

- under 5
- 5 – 10
- 10 – 25
- 25 – 50
- over 50



A Subset of the Dataset

ID	District name	<i>Y</i>	<i>E</i>	<i>SMR</i>	<i>X</i>	Neighbours
1	Skye-Lochalsh	9	1.38	6.52	16	5,9,11,19
2	Banff-Buchan	39	8.66	4.50	16	7,10
3	Caithness	11	3.04	3.62	10	6,12
11	Western Isles	13	4.40	2.95	7	1,5,9,12
15	NE Fife	17	7.84	2.17	7	25,29,50
17	Badenoch	2	1.07	1.87	10	7,9,13,16,19,29
26	Dunfermline	15	12.49	1.20	1	25,29,42,43
38	Monklands	8	9.35	0.86	1	30,42,44,49,51,54
42	Falkirk	8	15.78	0.51	16	26,30,34,38,43,51
45	Edinburgh	19	50.72	0.37	1	28,30,33,56
49	Glasgow	28	88.66	0.32	0	38,40,41,44,47,48,52,53,54
50	Dundee	6	19.62	0.31	1	15,21,29
55	Annandale	0	4.16	0	16	18,20,24,27,56
56	Tweeddale	0	1.76	0	10	18,24,30,33,45,55

A Hierarchical Bayesian Spatial Model for y_i 's

- **A model for the observed variables given latent variables**

$$y_i | X_i, E_i, \phi_i \sim \text{Poisson}(\exp(X_i \beta + \phi_i + \log(E_i))),$$

where X_i is a design vector, β is a vector of coefficients, ϕ_i is a spatial adjustment, and $\log(E_i)$ accounts for differences in expected values.

- **CAR priors for spatial random effects**

$$\phi_{1:n} \sim N(0, [\tau(D - \alpha W)^{-1}])$$

where τ is a spatially varying precision parameter, α is a parameter that controls spatial dependence ($|\alpha| < 1$), D is the diagonal matrix with the number of neighbours for location i , and W is the adjacency matrix.

- **The posterior distribution**

$$p(\phi_{1:n}, \beta, \alpha, \tau | y) \propto p(y | \phi_{1:n}, \beta) p(\phi_{1:n} | \alpha, \tau) p(\alpha) p(\tau) p(\beta)$$

The Need for Residual Diagnostics for Bayesian Models

- How do we know that a fitted model is adequate for the dataset? This is often referred to as a model diagnosis or model-checking problem.
- **Residual diagnostics** plays a very important role in frequentist statistics:
 - Assessing the normality of residuals with graphical tools, such as scatterplot, QQ plot, etc.
 - Using residual plots for discovering the problem of a fitted model, such as wrong distributional family, non-linear covariate effect, and non-constant variance.
 - Identifying outliers
 - Overall goodness-of-fit (GOF) and other statistical tests for quantifying the inadequacy of a fitted model

The Need for Residual Diagnostics for Bayesian Models (continued)

- However, to the best of our knowledge, residual diagnostics is not a common practice in Bayesian statistics due to the limitation of currently available tools.
 - Posterior predictive checks (PPC) are often used. However, the conservatism (lack of uniformity) of the PPC p-value due to the double use of a dataset is widely criticized.
 - A few methods have been proposed to remedy the conservatism problem in posterior predictive checks. However, they are computationally challenging.
 - More importantly, a single p-value does not answer all the questions that residual diagnostics can answer.
 - Can we have residual diagnostics for Bayesian models?

Section 2

Posterior Predictive Checks and the P-value with a χ^2 discrepancy function

Posterior Predictive Checks

- **Compares observed data (y) with simulated replicated data (y^{rep}) generated from the posterior predictive distribution:**

$$p(y^{\text{rep}} | y) = \int p(y^{\text{rep}} | \theta) p(\theta | y) d\theta,$$

- y^{rep} : Replicated data under the fitted model
- θ : Posterior parameters
- **Computation Steps:**
 - Simulate θ from the posterior distribution, $p(\theta|y)$
 - Simulate y^{rep} from the predictive distribution, $p(y^{\text{rep}}|\theta)$
 - Compare y to the replicated datasets y^{rep}
- **Posterior Predictive P-value using Discrepancy Functions**
 - Pearson χ^2 Discrepancy: $T(\mathbf{y}, \theta) = \sum_i \frac{(y_i - E(y_i|\theta))^2}{\text{Var}(y_i|\theta)}$
 - The Posterior Predictive P-value:
$$P = \frac{1}{m} \sum_{i=1}^m \mathbb{I}(T(\mathbf{y}, \theta_i) < T(\mathbf{y}_i^{\text{rep}}, \theta_i))$$

where $\theta_i \sim p(\theta|\mathbf{y})$, $\mathbf{y}_i^{\text{rep}} \sim p(\mathbf{y}|\theta_i)$, m is the number of posterior samples, $\mathbb{I}(\cdot)$ is the indicator function.

Section 3

Cross-validators Z-residuals Computed with Importance Sampling

Subsection 1

Z-residual Given the True Parameters

Randomized Predictive P-value Given the True Parameters

Given the true parameters θ and latent variable ϕ_i (eg, the spatial random effects), the randomized predictive p-value is defined as follows.

- Predictive p -value (PP) for continuous y_i is defined as follows:

$$pp(y_i^{\text{obs}}|\theta, \phi_i) = Pr(y_i > y_i^{\text{obs}}|\theta, \phi_i). \quad (1)$$

- The predictive p-value is also called probability integral transform (PIT) in the literature. The predictive p-value, [as a function of random \$y_i^{\text{obs}}\$](#) , is uniformly distributed on $(0, 1)$ for continuous y_i .
- When y_i is discrete, middle-value predictive p -value (MPP) is defined in the literature (Marshall and Spiegelhalter 2007 and others):

$$mpp(y_i^{\text{obs}}|\theta, \phi_i) = Pr(y_i > y_i^{\text{obs}}|\theta, \phi_i) + 0.5Pr(y_i = y_i^{\text{obs}}|\theta, \phi_i). \quad (2)$$

Randomized Predictive P-value Given the True Parameters (continued)

- Randomized predictive p -value (RPP) is defined as follows:

$$\text{rpp}(y_i^{\text{obs}}|\boldsymbol{\theta}, \phi_i) = \Pr(y_i > y_i^{\text{obs}}|\boldsymbol{\theta}, \phi_i) + U_i \Pr(y_i = y_i^{\text{obs}}|\boldsymbol{\theta}, \phi_i). \quad (3)$$

- The RPP is uniformly distributed on $(0, 1)$ under the true model with the true parameters for y_i is plugged in (3). Feng, Li, and Sadehpour (2020) provide a proof for this fact.
- The difference between MPP and RPP lies in the magnitude of $\Pr(y_i = y_i^{\text{obs}}|\boldsymbol{\theta}, \phi_i)$. For many applications with large y_i , this value is small. However, theoretically, we need the randomization to produce truly uniformly distributed predictive p -values.

Z-residuals Given the True Parameters

- We can transform $\text{rpp}(y_i^{\text{obs}}|\boldsymbol{\theta}, \phi_i)$ with the negative normal quantile:

$$z_i(y_i^{\text{obs}}|\boldsymbol{\theta}, \phi_i) = -\Phi^{-1}(\text{rpp}(y_i^{\text{obs}}|\boldsymbol{\theta}, \phi_i)). \quad (4)$$

where $\Phi()$ is the cumulative distribution function (CDF) of a standard normal distribution.

- When the model for y_i , including the parameters $\boldsymbol{\theta}$ and ϕ_i , is correctly specified, the $z_i(y_i^{\text{obs}}|\boldsymbol{\theta}, \phi_i)$ is distributed as the standard normal.
- We call the residuals defined in (4) by **Z-residuals** for the above reason.

Subsection 2

Posterior Z-residual

Posterior Z-residual

- The **full data posterior** density of $(\phi_{1:n}, \theta)$ given observations $\mathbf{y}_{1:n}^{\text{obs}}$ is given by:

$$P_{\text{post}}(\theta, \phi_{1:n} | \mathbf{y}_{1:n}^{\text{obs}}) = \prod_{j=1}^n P_Y(y_j^{\text{obs}} | \phi_{1:n}, \theta) P_{\phi}(\phi_{1:n} | \theta) \pi(\theta) / C_1. \quad (5)$$

- The posterior predictive p -value:

$$\text{rpp}^{\text{Post}}(y_i^{\text{obs}}) = E_{\text{post}}(\text{rpp}(y_i^{\text{obs}} | \theta, \phi_i)), \quad (6)$$

- $\text{rpp}(y_i^{\text{obs}} | \theta, \phi_i)$ is from equation (3).
- Given MCMC samples $\{(\theta^{(t)}, \phi_{1:n}^{(t)}); t = 1, \dots, T\}$ from the full data posterior (5), posterior predictive p -value (6) is computed as follows:

$$\widehat{\text{rpp}}^{\text{Post}}(y_i^{\text{obs}}) = \frac{\sum_t^T \text{rpp}(y_i^{\text{obs}} | \theta^{(t)}, \phi_i^{(t)})}{T}. \quad (7)$$

Posterior Z-residual (continued)

- It is preferred to transfer $\text{rpp}(y_i^{\text{obs}}|\boldsymbol{\theta}, \phi_i)$ with the negative normal quantile:

$$z_i^{\text{Post}}(y_i^{\text{obs}}) = -\Phi^{-1}(\widehat{\text{rpp}}^{\text{Post}}(y_i^{\text{obs}})). \quad (8)$$

where $\Phi()$ is the cumulative distribution function (CDF) of a standard normal distribution. We call the residuals in (8) by **Posterior Z-residuals**.

- Our simulation studies will illustrate the conservatism of posterior Z-residuals.

Subsection 3

Importance-sampling Cross-validators Z-residual

Cross-validators (CV) Z-residual

- CV distribution $P_{\text{post}(-i)}(\boldsymbol{\theta}, \phi_{1:n} | \mathbf{y}_{-i}^{\text{obs}})$ is given as follows:

$$P_{\text{post}(-i)}(\boldsymbol{\theta}, \phi_{1:n} | \mathbf{y}_{-i}^{\text{obs}}) = \prod_{j \neq i} \frac{P_y(y_j^{\text{obs}} | \phi_j, \boldsymbol{\theta}) P_\phi(\phi_{1:n} | \boldsymbol{\theta}) \pi(\boldsymbol{\theta})}{C_2}. \quad (9)$$

- The CV predictive p -value for y_i^{obs}

$$\text{rpp}(y_i^{\text{obs}} | \mathbf{y}_{-i}^{\text{obs}}) = E_{\text{post}(-i)}(\text{rpp}(y_i^{\text{obs}} | \boldsymbol{\theta}, \phi_i)). \quad (10)$$

- Given MCMC samples $\{(\boldsymbol{\theta}^{(t)}, \phi_{1:n}^{(t)}); t = 1, \dots, T\}$ from the CV posterior (9), the CV predictive p -value is computed as follows:

$$\widehat{\text{rpp}}^{\text{CV}}(y_i^{\text{obs}}) = \frac{\sum_{t=1}^T \text{rpp}(y_i^{\text{obs}} | \boldsymbol{\theta}^{(t)}, \phi_i^{(t)})}{T}. \quad (11)$$

- The Z-residual for y_i^{obs} is given by:

$$z_i^{\text{CV}}(y_i^{\text{obs}}) = -\Phi^{-1}(\widehat{\text{rpp}}^{\text{CV}}(y_i^{\text{obs}})). \quad (12)$$

- MCMC sampling needs to be redone for calculating each $z_i^{\text{CV}}(y_i^{\text{obs}})$.

Review of Importance Sampling

- Our goal is to find the expectation of a function $a(X)$ when X has a probability density proportional to $f(x)$, denoted by $E_f(a(X))$
- Instead of drawing samples from $f(x)$, we draw samples from an approximate distribution with a probability density $g(x)$.
- Importance weight: $w(x) = \frac{f(x)}{g(x)}$
- *Importance reweighting formula*

$$E_f(a(X)) = \frac{E_g(a(X)w(X))}{E_g(w(X))}. \quad (13)$$

- The intuitive explanation of the *importance reweighting formula* (13) is that samples that are more compatible with the target distribution f will be assigned more weight (and vice versa).

Importance-Sampling CV (ISCV) Z-residual

- We can estimate expectations with respect to $P_{\text{post}(-i)}(\boldsymbol{\theta}, \boldsymbol{\phi}_{1:n} | \mathbf{y}_{-i}^{\text{obs}})$ in (9) by reweighting samples from $P_{\text{post}}(\boldsymbol{\theta}, \boldsymbol{\phi}_{1:n} | \mathbf{y}_{1:n}^{\text{obs}})$ in (5):

$$\text{rpp}(y_i^{\text{obs}} | \mathbf{y}_{-i}^{\text{obs}}) = \frac{E_{\text{post}}[\text{rpp}(y_i^{\text{obs}} | \boldsymbol{\theta}, \boldsymbol{\phi}_i) w_i^{\text{IS}}(\boldsymbol{\theta}, \boldsymbol{\phi}_i)]}{E_{\text{post}}[w_i^{\text{IS}}(\boldsymbol{\theta}, \boldsymbol{\phi}_i)]}, \quad (14)$$

where,

$\text{rpp}(y_i^{\text{obs}} | \boldsymbol{\theta}, \boldsymbol{\phi}_i)$ is from equation (3), and

$w_i^{\text{IS}}(\boldsymbol{\theta}, \boldsymbol{\phi}_i)$ is the ratio:

$$w_i^{\text{IS}}(\boldsymbol{\theta}, \boldsymbol{\phi}_i) = \frac{P_{\text{post}(-i)}(\boldsymbol{\theta}, \boldsymbol{\phi}_{1:n} | \mathbf{y}_{-i}^{\text{obs}})}{P_{\text{post}}(\boldsymbol{\theta}, \boldsymbol{\phi}_{1:n} | \mathbf{y}_{1:n}^{\text{obs}})} \times \frac{C_2}{C_1} = \frac{1}{P_y(y_i^{\text{obs}} | \boldsymbol{\theta}, \boldsymbol{\phi}_i)}. \quad (15)$$

Importance-Sampling CV (ISCV) Z-residual (continued)

- Given MCMC samples $\{(\boldsymbol{\theta}^{(t)}, \phi_{1:n}^{(t)}); t = 1, \dots, T\}$ from the full data posterior (5), the IS predictive p -value (14) is computed as follows:

$$\widehat{\text{rpp}}^{\text{ISCV}}(y_i^{\text{obs}}) = \frac{\sum_{t=1}^T \left[\text{rpp}(y_i^{\text{obs}} | \boldsymbol{\theta}^{(t)}, \phi_i^{(t)}) w_i^{\text{IS}}(\boldsymbol{\theta}^{(t)}, \phi_i^{(t)}) \right] / T}{\sum_t^T w_i^{\text{IS}}(\boldsymbol{\theta}^{(t)}, \phi_i^{(t)}) / T}. \quad (16)$$

- We then transfer $\widehat{\text{rpp}}^{\text{ISCV}}(y_i^{\text{obs}})$ with the negative normal quantile:

$$z_i^{\text{ISCV}}(y_i^{\text{obs}}) = -\Phi^{-1}(\widehat{\text{rpp}}^{\text{ISCV}}(y_i^{\text{obs}})). \quad (17)$$

- We call the residuals in (17) by **Importance-Sampling Cross-validatory (ISCV) Z-residuals**.

Section 4

Simulation Study I: Detecting Wrong Family

Subsection 1

Data Generating and Model Fitting

- A covariate $X_i \sim \text{Uniform}(0, 30)$,
- Adjacency matrix $W \sim \text{Erdős-Rényi model } G(n, p)$, where n is the number of districts and $p = 0.1$ is the probability for drawing an edge between two arbitrary districts.
- Precision matrix is calculated by $\tau(D - \alpha W)$, where $\tau = 0.5$, $\alpha = 0.5$, and D is an $n \times n$ diagonal matrix with the number of neighbours for each district.
- The random vector $\phi_{1:n}$ is generated from the CAR prior

$$\phi_{1:n} \sim N(0, \tau(D - \alpha W)^{-1}) \quad (18)$$

- Expected cases: $E_i \sim \text{Poisson}(10)$

- The count response

$$y_i \sim NB(\mu_i = \exp(0.01X_i + \phi + \log(E_i)), \gamma) \quad (19)$$

where $\gamma = 2.5$ is the parameter that controls the overdispersion of NB.

- We generated datasets with sample size $n = 50, 100, 200$. For each sample size n , we generated 200 datasets for estimating model rejection rates of different diagnostics methods.

Model Fitting

- **True model:** $y_i \sim \text{NB}(\exp(X_i\beta + \phi_i + \log(E_i)), \gamma)$
- **Wrong model:** $y_i \sim \text{Poisson}(\exp(X_i\beta + \phi_+ \log(E_i)))$
- Priors for fitting models are specified as follows:
 - $\phi_1, \dots, \phi_n \sim \text{MultiNormal}(0, [\tau(D - \alpha W)]^{-1})$.
 - $\beta \sim \text{N}(0, 1000)$
 - $\tau \sim \text{Gamma}(0.001, 0.001)$
 - $\alpha \sim \text{Uniform}(0, 1)$
 - $\gamma \sim \text{Gamma}(0.001, 0.001)$ (Only for the true model)
- We used rstan to fit these models.

Goals of Simulation Studies

- We calculated four different kinds of Z-residual and compared them:
 - Based on middle-value predictive p-value (MPP) :

$$\text{Post. mpp Z-residual: } z_i^{\text{Post}}(y_i^{\text{obs}}) = -\Phi^{-1}(\widehat{\text{mpp}}^{\text{Post}}(y_i^{\text{obs}})) \quad (20)$$

$$\text{ISCV mpp Z-residual: } z_i^{\text{ISCV}}(y_i^{\text{obs}}) = -\Phi^{-1}(\widehat{\text{mpp}}^{\text{ISCV}}(y_i^{\text{obs}})) \quad (21)$$

- Based on randomized predictive p-values (RPP):

$$\text{Post. rpp Z-residual: } z_i^{\text{Post}}(y_i^{\text{obs}}) = -\Phi^{-1}(\widehat{\text{rpp}}^{\text{Post}}(y_i^{\text{obs}})) \quad (22)$$

$$\text{ISCV rpp Z-residual: } z_i^{\text{ISCV}}(y_i^{\text{obs}}) = -\Phi^{-1}(\widehat{\text{rpp}}^{\text{ISCV}}(y_i^{\text{obs}})) \quad (23)$$

- Our simulation studies will focus on comparing the difference between these four types of Z-residuals and the PP check, demonstrating the effectiveness of ISCV Z-residuals in discerning true and wrong models.

Subsection 2

Z-residual Diagnostics for A Single Dataset

Posterior V.S. ISCV Predictive P-value: Histogram

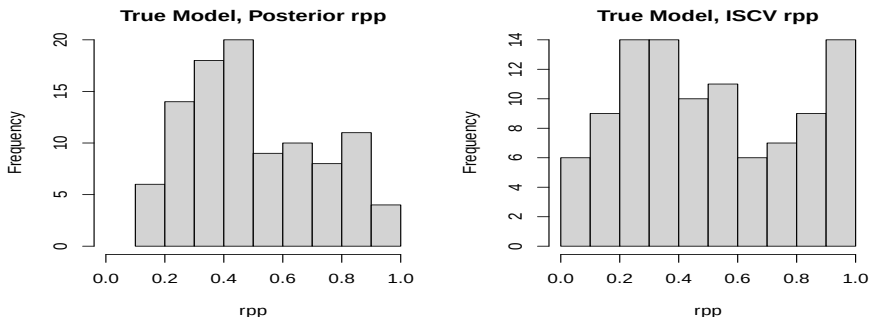


Figure 1: The true model is an NB model. The sample size is $n = 100$.

- The posterior RPPs are more concentrated about 0.5 than truly uniform distribution on $(0, 1)$; the ISCV RPPs are more spread-out on $(0, 1)$ with more p-values closer to 0 and 1.
- Correspondingly, the posterior Z-residuals are more concentrated about 0 with a standard deviation smaller than 1. The shape of the distribution of the posterior Z-residuals may also deviate from the standard normal.

Posterior Z-residual: Scatterplot

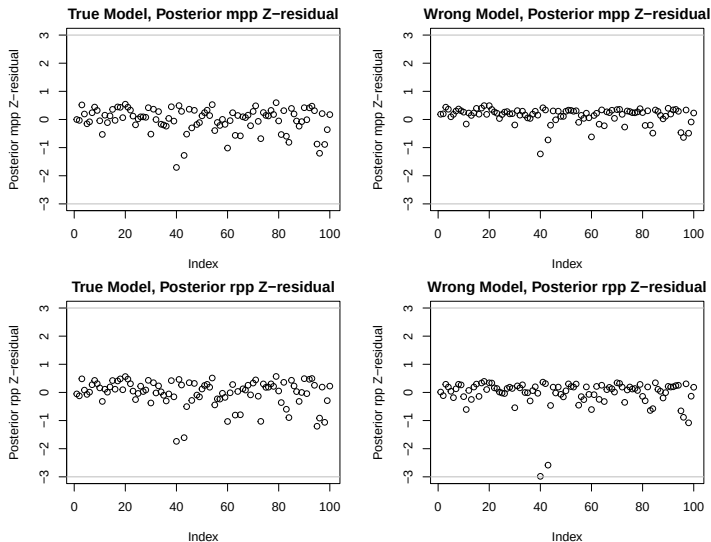


Figure 2: The true model is an NB model. The wrong model is a Poisson model. The sample size is $n = 100$.

Posterior Z-residual: QQ Plot

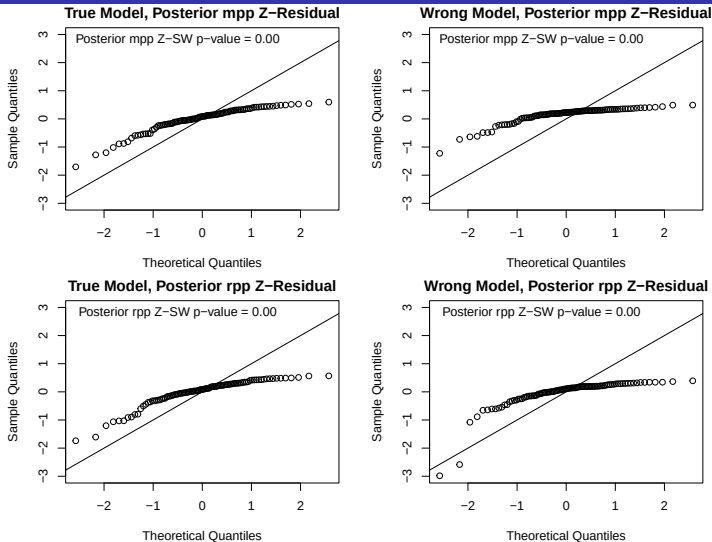


Figure 3: The true model is an NB model. The wrong model is a Poisson model. The sample size is $n = 100$.

ISCV Z-residual: Scatterplot

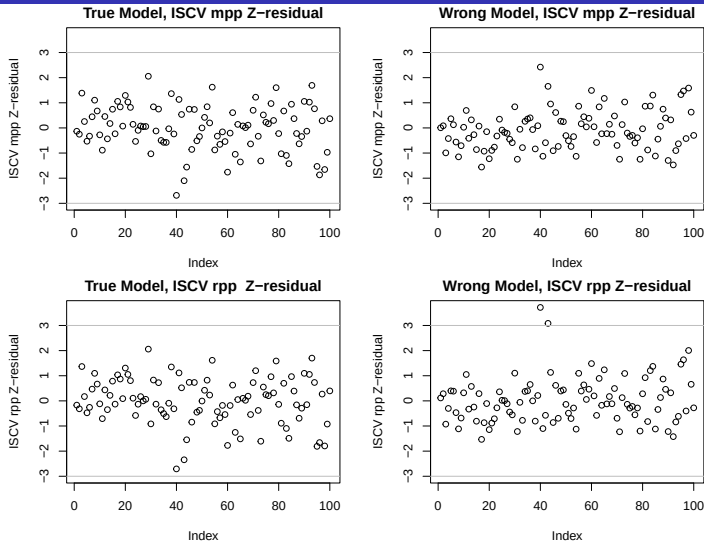


Figure 4: The true model is an NB model. The wrong model is a Poisson model. The sample size is $n = 100$.

ISCV Z-residual: QQ Plot

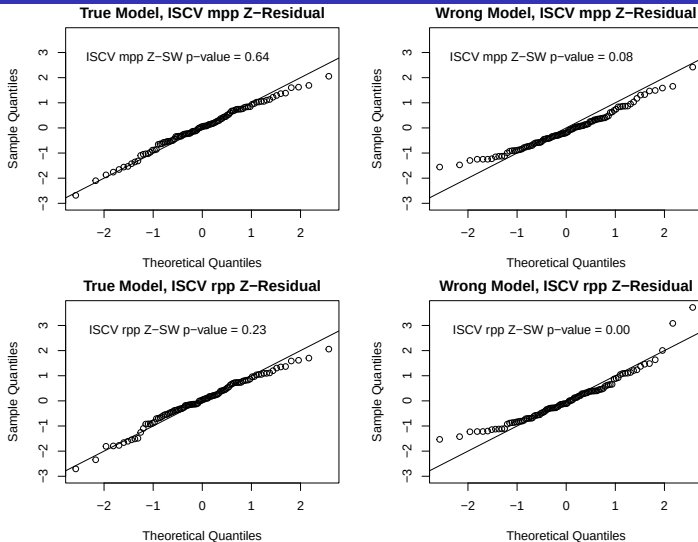


Figure 5: The true model is an NB model. The wrong model is a Poisson model. The sample size is $n = 100$.

Subsection 3

Type-I Error Rates and Powers of SW Tests based on Z-residuals

The histogram of the P-values

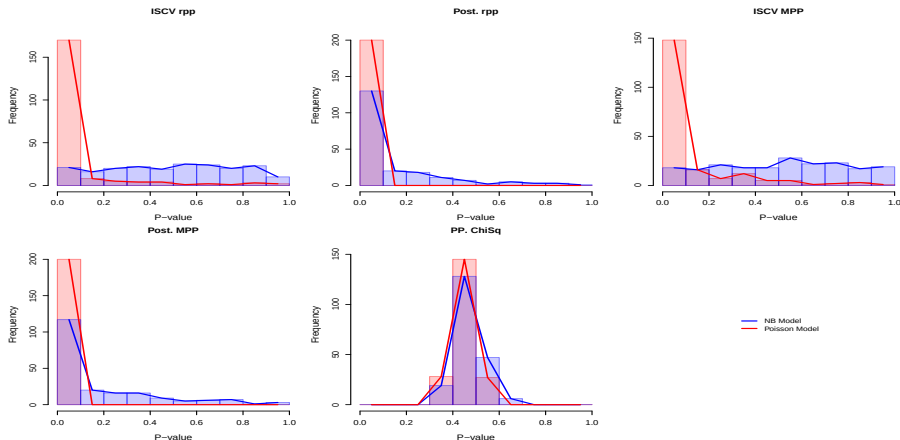


Figure 6: Histograms of P-values from Z-residual-based Shapiro-Wilk (SW) tests and posterior predictive χ^2 tests. Results are based on 200 replications for a sample size of $N=100$.

Model Rejection Rates of Z-residual-based SW Tests

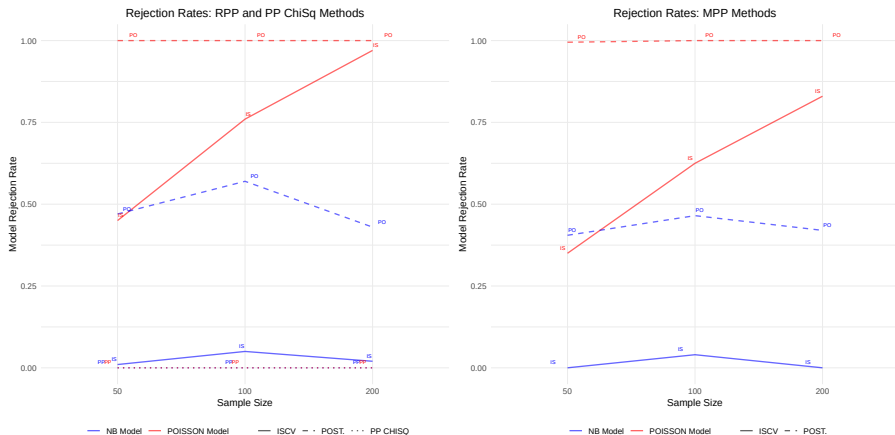


Figure 7: Model rejection rates of Z-residual-based Shapiro-Wilk (SW) tests and posterior predictive χ^2 tests. A model is rejected when the test p-value is smaller than 5%. For all investigated sample sizes (N=50, 100, 200), 200 replications were performed.

Section 5

Simulation Study II: Detecting Non-linear Covariate Effects

Subsection 1

Data Generating and Model Fitting

Data Generation and Model Fitting

- The count dataset is generated from a negative binomial distribution with mean

$$\mu_i = \exp(0.05X_i^2 + \phi_i + \log(E_i))$$

and the parameter $\gamma=2.5$ controls the NB overdispersion, then $y_i \sim NB(\mu_i, \gamma)$.

- The other parameter setting remains the same as used in generating datasets for investigating the detection of the mis-specification of family.
- Model Fitting
 - **True model:** $y_i \sim NB(\exp(X_i^2\beta + \phi_i + \log(E_i)), \gamma)$
 - **Wrong model:** $y_i \sim NB(\exp(X_i\beta + \phi_i + \log(E_i)), \gamma)$
 - The priors as follows are specified for fitting models:
 - $\phi_{1:n} \sim \text{MultiNormal}(0, [\tau(D - \alpha W)]^{-1})$.
 - $\beta \sim N(0, 1000)$
 - $\tau \sim \text{gamma}(0.001, 0.001)$
 - $\alpha \sim \text{Uniform}(0, 1)$
 - $\gamma \sim \text{gamma}(0.001, 0.001)$

Subsection 2

Z-residual Diagnostics for A Single Dataset

Posterior Z-residual: Scatterplot

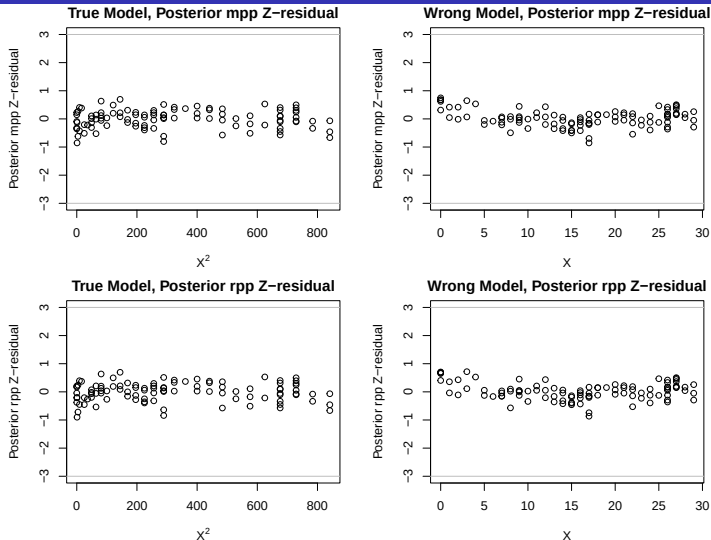


Figure 8: The true model is an NB model with X^2 as a covariate. The wrong model is an NB model with X as a covariate. The sample size is $n = 100$.

Posterior Z-residual: Boxplot

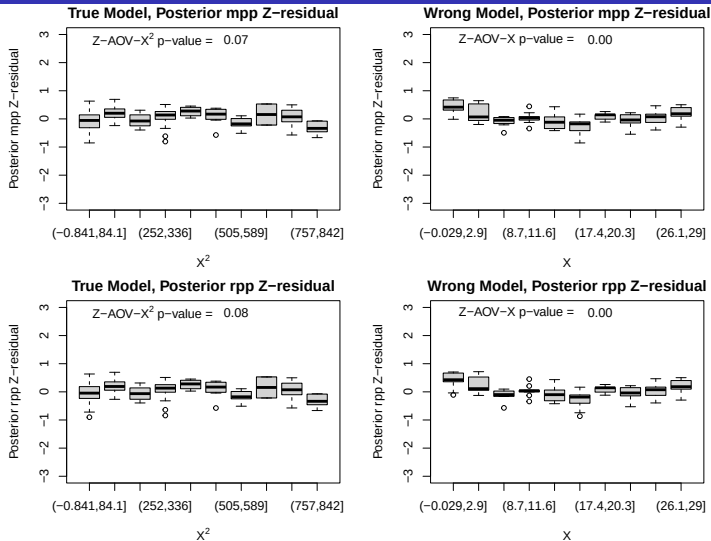


Figure 9: The true model is an NB model with X^2 as a covariate. The wrong model is an NB model with X as a covariate. The sample size is $n = 100$.

ISCV Z-residual: Scatterplot

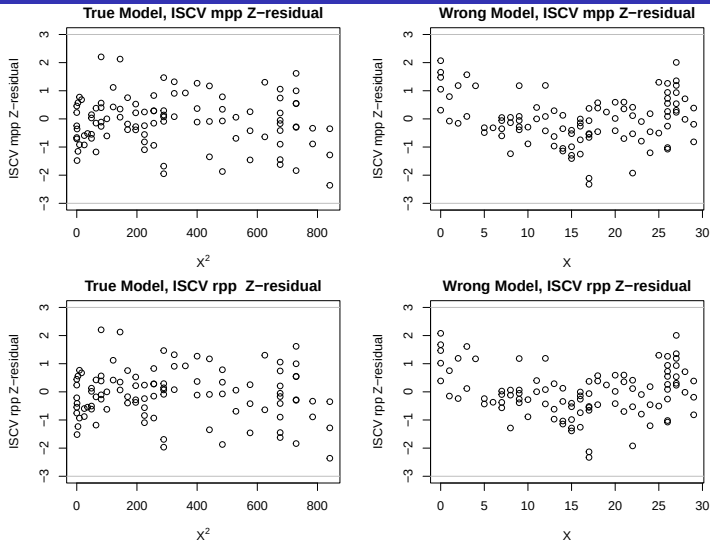


Figure 10: The true model is an NB model with X^2 as a covariate. The wrong model is an NB model with X as a covariate. The sample size is $n = 100$.

ISCV Z-residual: Boxplot

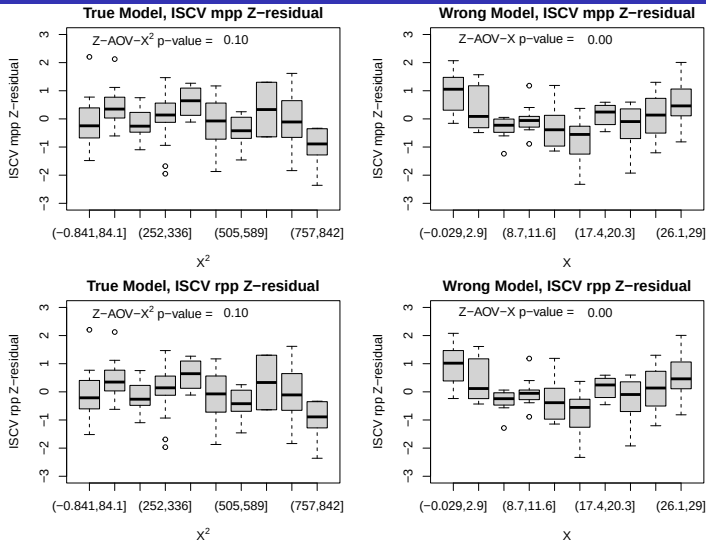


Figure 11: The true model is an NB model with X^2 as a covariate. The wrong model is an NB model with X as a covariate. The sample size is $n = 100$.

Subsection 3

Type-I Error Rates and Powers of ANOVA Tests based on Z-residuals

Model Rejection Rates of Z-residual-based ANOVA Tests

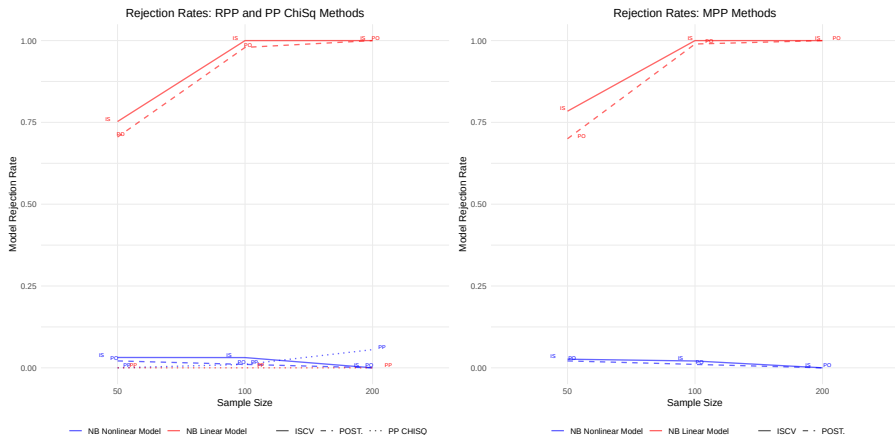


Figure 12: Model rejection rates of Z-residual-based ANOVA F-tests and posterior predictive χ^2 tests. A model is rejected when the test p-value is smaller than 5%. For all investigated sample sizes (N=50, 100, 200), 100 replications were performed.

Section 6

Conclusions and Future Work

Conclusions and Future Work

- The ISCV Z-residuals defined with randomized predictive p-values work well for checking Bayesian models. They can be used to diagnose Bayesian models in almost the same way that Pearson's residuals are used for frequentist regression models, including graphical diagnosis, statistical tests, and outlier detection.
- The replicated studies show that the ISCV Z-residuals have expected frequentist properties: the type-I errors are close to the nominal level and there are powers when the model is mis-specified.
- We will compare this tool in terms of type-I error rates and powers with other predictive checks' p-values (eg, pivotal quantities (Johnson 2007), and possibly others). We will also apply it to other Bayesian models.

References

- Wu, T., Li, L., Feng, C. (2024). Z-residual diagnostic tool for assessing covariate functional form in shared frailty models. Journal of Applied Statistics 52(1), 28–58.
- Feng, C., Li, L., and Sadeghpour, A. (2020). A comparison of residual diagnosis tools for diagnosing regression models for count data. BMC Medical Research Methodology. 20(1):175.
- Vehtari, A., Gelman A., and Gabry, J. (2017) Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC. Statistics and Computing 27(5):1413-32.
- Li, L., Feng, C.X., and Qiu, S.* (2017), Estimating Cross-validatory Predictive p -values with Integrated Importance Sampling for Disease Mapping Models. Statistics in Medicine, Volume 36, Issue 14, Pages 2220-2236.
- Li, L., Qiu, S.*, Zhang, B.*, and Feng, C.X. (2016). Approximating Cross-validatory Predictive Evaluation in Bayesian Latent Variables Models with Integrated IS and WAIC. Statistics and Computing, Volume 26, Issue 4, pp 881-897.